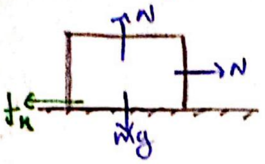
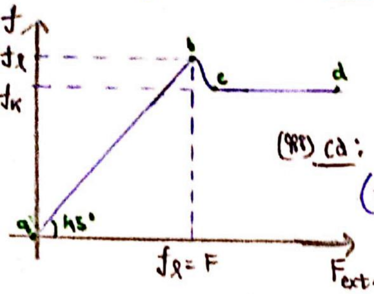
Friction

Variation of Friction with External Force:



(i) ab: static friction (block is at rest w.r.t. ground)

(ii) At b: $(f_s)_{max}$ or f_l



(iii) cd: kinetic friction ($f_K < (f_s)_{max}$)

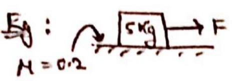
Generally

$\boxed{\mu_s = \mu_K = \mu}$

(iv) $(f_s)_{max} = \mu_s N$

$f_K = \mu_K N$

Eg:



(i) When F = 8N

(ii) When F = 20N

Sol^n (i) $N = 50N$

$(f_s)_{max} = \mu \cdot N = 0.2 \times 50 = 10N$

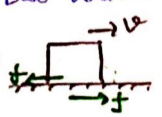
So, when F = 8N, $\boxed{f = 8N}$

(ii) When F = 20N $\Rightarrow \boxed{f = 10N}$ Ans

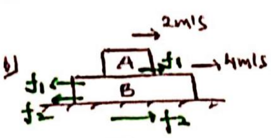
Direction of Friction:

→ Friction opposes relative motion between two bodies.

a)

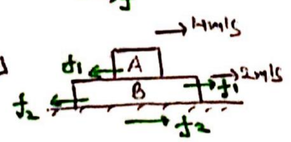


b)



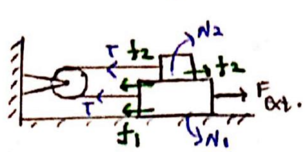
$v_{A/B} = 2-4 = -2\ m/s$

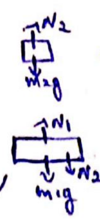
c)



$v_{A/B} = 4-2 = 2\ m/s$

d)





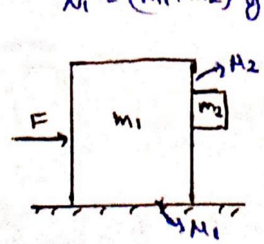
$N_2 = m_2 g$

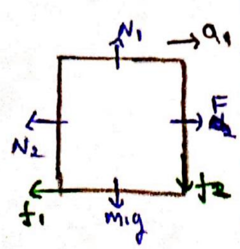
$N_1 = m_1g + N_2$

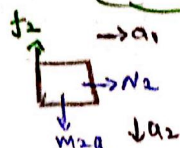
$N_1 = m_1g + m_2g$

$N_1 = (m_1 + m_2)g$

e)







For m_1

$F - N_2 - f_1 = m_1 a_1$ —(1)

$f_1 = \mu_1 N_1$ —(2)

$N_1 = m_1 g + f_2$ —(3)

For m_2 Slipping

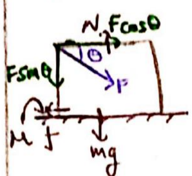
$\boxed{f_2 = \mu_2 N_2}$

where, $N_2 = m_2 a_1$

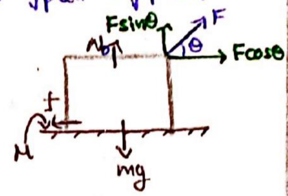
$m_2 g - f_2 = m_2 a_2$

Pushing and Pulling:

→ Pulling is easier than pushing because $N_{pull} < N_{push} \Rightarrow f_{pull} < f_{push}$



(Pushing)



(Pulling)

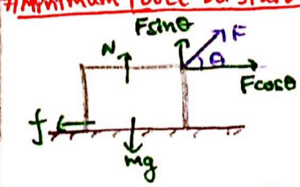
$N = F\sin\theta + mg$

$f = \mu N$

$N_0 = mg - F\sin\theta$

$f = \mu N_0$

Minimum Force to start slipping:



To just start slipping ⇒

$F\cos\theta = f = \mu N$

$\Rightarrow F\cos\theta = \mu(mg - F\sin\theta)$

$\Rightarrow F = \dfrac{\mu mg}{\cos\theta + \mu\sin\theta}$

To minimise F, denominator should be maximized ⇒

$\frac{d}{d\theta}(\cos\theta + \mu\sin\theta) = 0$

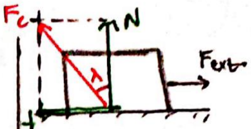
$\Rightarrow \boxed{\tan\theta = \mu}$

Also, $\boxed{F_{min.} = \dfrac{\mu mg}{\sqrt{\mu^2+1}}}$

Contact ~~Angle~~ Force: (F_c)

→ Contact force is the resultant force of normal force and frictional force acting on a body.

ie $\boxed{\vec{F_c} = \vec{f} + \vec{N}}$



$F_c = \sqrt{N^2 + f^2}$

$(F_c)_{min} = N\ (f = 0)$

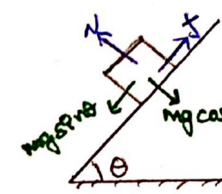
$(F_c)_{max} = N\sqrt{1+\mu^2}$

$\Rightarrow \boxed{N \le F_c \le N\sqrt{1+\mu^2}}$

(In case of μ_K & μ_s is given, take use of μ_s for above relation)

Angle of Repose:

→ Minimum angle for which body slips due to its weight on rough inclined surface.



→ Block would start to just slip if

$mg\sin\theta = \mu N$

$\Rightarrow mg\sin\theta = \mu mg\cos\theta$

$\Rightarrow \boxed{\theta = \tan^{-1}(\mu)}$

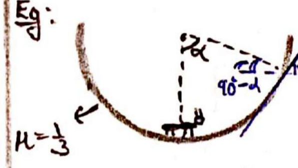
Angle of Repose

If $\theta < \tan^{-1}(\mu) \Rightarrow$ No slipping

If $\theta > \tan^{-1}(\mu) \Rightarrow$ slipping

↳ $(a = g\sin\theta - \mu g\cos\theta)$

Eg:



$\mu = \frac{1}{3}$

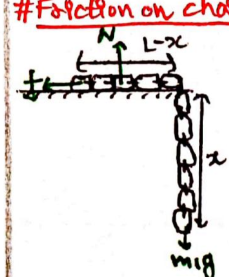
Find max. possible α upto which insect can crawl w/o slipping.

Sol^n At limiting situation

$\tan\alpha = \mu \Rightarrow \frac{1}{3}$

$\cot\alpha = 3 \Rightarrow \boxed{\alpha = \cot^{-1}(3)}$ Ans

Friction on chain:



$m_1 \to$ Hanging mass.

→ For maximum length without slipping

$m_1 g = f_{max}$

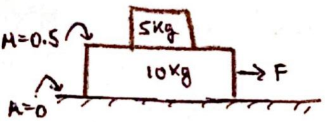
$\frac{m}{L} \cdot x \cdot g = \mu \frac{m}{L}(L-x)g$ (N)

$\Rightarrow x = \mu(L-x)$

$\boxed{x = \dfrac{\mu L}{1+\mu}}$

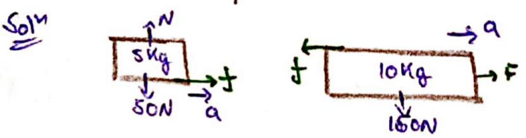
Two Block System :

1) External Force on lower Block ⇒



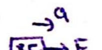
(i) Find max. force upto which both blocks move with same acc^n / 5kg block does not slip.

Sol^n



Since both blocks are considered to be moving with same acc^n, They can be treated as a single body.

Hence



$F = 15a$

$\Rightarrow \boxed{a = \frac{F}{15}}$

For 5kg Block

$f = 5a \Rightarrow 5 \times \frac{F}{15}$

$\boxed{f = \frac{F}{3}}$

The max. value of force would be

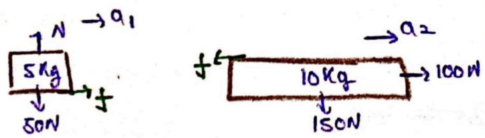
$f \leq f_{max}$

$\frac{F}{3} \leq \mu N \Rightarrow \frac{F}{3} \leq 0.5 \times 50$

$\boxed{F \leq 75N} \Rightarrow \boxed{F_{max} = 75N}$ Ans

(ii) What if $F = 100N$?

Sol^n In such case, both blocks will have diff. acc^n & they will not move together.

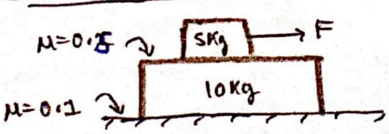


$\boxed{a_1 = \frac{25}{5} = 5 m/s^2}$ (since limiting friction)

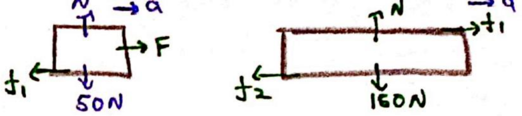
$100 - 25 = 10a_2$

$\Rightarrow \boxed{a_2 = 7.5 m/s^2}$

2) External Force on upper Block ⇒



(i) Find max. force upto which both blocks move together.



When F is max, the friction would also be max.

$f_{1,max} = 0.5 \times 50 \Rightarrow 25 N$

$f_{2,max} = 0.1 \times 150 \Rightarrow 15 N$

For 10kg block

$f_2 - f_1 = -10a$ | $15 - 25 = -10a$

$\Rightarrow \boxed{a = 1 m/s^2}$

For 5kg block

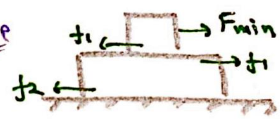
$F_{max} - f_1 = 5a$ | $F_{max} - 25 = 5(1)$

$\boxed{F_{max} = 30N}$ Ans

(ii) What would be F_{min} at which both blocks will just start moving together?

Sol^n For minimum force, the friction acting on upper block should be just enough to overcome the friction b/w 10kg block & ground.

i.e



$f_1 = f_2$

And in order to move 10kg block, f_2 should be max.

$f_{2,max} = 15N$

So, $f_1 = 15N \Rightarrow \boxed{F_{min} = 15N}$

Hence, range of external force, for which both blocks move together with same acc^n is

$\boxed{15N \text{ to } 30N}$